Cognitive–Cultural Fit Model (CCFM):
An Integrative Framework for Person–Organization Alignment
Based on Cognitive Science and Artificial Intelligence

David Suárez
Independent Researcher
Ecuador
Email: suarezdavid6824@gmail.com
Date: February 2026

Abstract

The assessment of person–organization fit (P–O fit) constitutes a central challenge in organizational psychology and applied cognitive science. Although organizational culture largely shapes the identity, cognitive patterns, and internal dynamics of institutions, selection and development processes often prioritize generic variables such as soft skills or personality traits. As a result, highly technically competent individuals may experience persistent friction in culturally misaligned environments—not due to individual deficiencies, but due to mismatches between cognitive and organizational identities.

This paper introduces the Cognitive–Cultural Fit Model (CCFM), a hybrid, multi-level evaluation framework that integrates principles from cognitive science, psychometrics, and artificial intelligence to analyze alignment between individuals and organizations across six cognitive–cultural dimensions. The model combines structured data derived from psychometric instruments with narrative information processed through natural language processing (NLP) and large language models (LLMs), using a weighted fusion mechanism that preserves interpretability, analytical traceability, and methodological rigor.

Through this integration, the system models cognitive attributes, observable cultural behaviors, and socioemotional competencies, generating standardized latent scores,

explanatory diagnostics, contextualized evidence, and actionable recommendations for selection, onboarding, role design, and cultural risk monitoring. Unlike approaches focused solely on classification or candidate filtering, the CCFM emphasizes understanding individuals' potential for adaptation and development, highlighting that performance emerges from contextual compatibility rather than absolute hierarchies of value.

A functional prototype and proof of concept are presented to illustrate the technical feasibility and analytical value of the proposed approach. The results suggest that integrating cognitive modeling, semantic analysis, and psychometric evaluation enables the development of more precise, transparent, and human-centered organizational diagnostic systems. The CCFM thus provides a scalable and extensible foundation for future research in AI-assisted evaluation and decision-making in complex organizational contexts.

Keywords: cognitive science, person–organization fit, natural language processing, organizational culture, artificial intelligence, HR analytics

1. Introduction

The relationship between individuals and organizations is mediated by complex systems of values, norms, expectations, and cognitive patterns that shape institutional culture (Schein, 2010; Hofstede, 2001). From the perspective of cognitive science and organizational psychology, such culture not only regulates observable behavior but also structures how individuals interpret information, make decisions, and define what constitutes adequate performance (Barsalou, 2008; Clark, 1997). In this sense, job performance emerges from processes of situated cognition, in which individual mental frameworks continuously interact with specific organizational contexts (Brown et al., 1989).

In practice, however, dominant paradigms in selection, evaluation, and professional development tend to overlook cognitive–cultural dimensions, focusing instead on more

general variables such as generic competencies or personality traits (Kristof, 1996; DeVellis, 2016). Moreover, many classical models of organizational culture assessment rely heavily on self-report measures, which are often affected by social desirability and strategic response bias (Podsakoff et al., 2003). These approaches also tend to favor aggregated and relatively static representations of culture, based on group averages and cross-sectional measurements (Denison, 1990). As a result, they exhibit limitations in capturing the semantic richness, contextual nuance, and dynamic nature of everyday language, where cognitive patterns, reasoning styles, and motivational orientations are expressed in ways that are difficult to consciously manipulate.

Consequently, situations of low performance, demotivation, or early turnover are often mistakenly attributed to individual deficits, when they in fact reflect broader contextual mismatches between individuals and the organizational environments in which they operate (Kristof-Brown et al., 2005).

Contemporary literature on person–organization fit (P–O fit) has consistently documented the influence of cultural alignment on variables such as commitment, job satisfaction, and performance (Kristof, 1996; Cable & Judge, 1996). However, many of these approaches rely predominantly on self-reported instruments and standardized surveys, with limited integration of behavioral, linguistic, and contextual data. As a result, they often provide general descriptions of cultural fit but lack explanatory and prescriptive mechanisms to meaningfully guide processes of individual adaptation and development within organizations.

In parallel, recent advances in natural language processing (NLP) and large language models (LLMs) have significantly expanded the possibilities for semantic and contextual analysis in organizational settings (Devlin et al., 2019; Brown et al., 2020; Bommasani et al.,

2021). These technologies enable the analysis of narratives, experiences, and communication styles with an unprecedented level of detail, as well as the identification of latent behavioral indicators associated with situated cognitive processes. However, their integration into human evaluation systems remains limited, fragmented, and, in many cases, methodologically opaque (Doshi-Velez & Kim, 2017).

In response to this gap, the present work introduces the Cognitive–Cultural Fit Model (CCFM), an integrative framework that combines principles from cognitive science, psychometrics, and artificial intelligence for the systematic analysis of person–organization alignment. The model is structured around six cognitive–cultural dimensions—goal orientation, identity and decision styles, emotional regulation, organizational preferences, judgment frameworks, and problem-processing styles—drawing inspiration from the multi-focus model of organizational culture (Hofstede et al., 2010).

Unlike purely descriptive or predictive approaches, the CCFM incorporates a controlled fusion mechanism that systematically and transparently integrates psychometric measurements, semantic analysis, and inferential signals derived from language models (Embretson & Reise, 2000; Ribeiro et al., 2016). This approach enables the triangulation of multiple sources of information, enhancing both statistical robustness and theoretical interpretability.

Through this integration, the system generates standardized scores, explanatory diagnostics, contextualized evidence, and actionable recommendations for selection, onboarding, role design, and cultural risk monitoring. In this way, the CCFM is positioned as both an explanatory and prescriptive model, oriented toward human development rather than the mere exclusion of candidates.

The remainder of this article is structured as follows: Section 2 presents the theoretical and methodological foundations of the model; Section 3 describes the system architecture and data integration mechanisms; Section 4 introduces a proof of concept; and Section 5 discusses implications, limitations, and future research directions.

2. Methods

2.1. Research Design

The present study adopts a mixed-methods, exploratory–applied research design aimed at the development and preliminary validation of a computational system for assessing cognitive–cultural alignment between individuals and organizations. The design integrates quantitative and qualitative approaches through the combination of structured psychometric instruments, automated natural language processing, and hybrid inference models.

From a mixed-methods perspective (Creswell, 2014), the design approximates a convergent triangulation model, in which quantitative and qualitative data sources are analyzed in parallel and integrated at a subsequent inference stage. This strategy enables the systematic comparison, complementarity, and validation of multiple layers of psychological, behavioral, and linguistic information.

Within this framework, the Cognitive–Cultural Fit Model (CCFM) is conceptualized as a multilevel evaluation system, in which self-reported data, open-ended narratives, and latent linguistic signals converge to capture cognitive, behavioral, and socioemotional dimensions of person–organization fit. This approach aligns with recent work highlighting the potential of multifocal models and artificial intelligence techniques for the dynamic analysis of organizational culture (Schachner et al., 2024).

2.2. Participants and Data Sources

Given the methodological and proof-of-concept nature of this study, the data analyzed consist of synthetic and simulated profiles, designed to represent realistic patterns of organizational interaction, cognitive styles, and professional trajectories.

This strategy enabled the control of contextual variables, the exploration of multiple alignment and misalignment scenarios, and the evaluation of system behavior without compromising sensitive personal information, in line with best practices for the early-stage development of computational assessment systems (Creswell, 2014).

The data sources included:

i) Responses to structured psychometric instruments.

ii) Open-ended narratives elicited through situational prompts.

iii) Simulated organizational profiles derived from institutional descriptions.

This approach facilitates initial model evaluation and establishes a foundation for future empirical validation using real-world samples.

2.3. Instruments

The CCFM integrates two complementary instruments designed to capture different layers of psychological, cognitive, and behavioral information.

2.3.1. Psychometric Instrument

The psychometric instrument consists of a structured battery of items with graded response scales, aimed at estimating latent traits associated with cognitive dispositions,

behavioral preferences, and socioemotional styles (Embretson & Reise, 2000; DeVellis, 2016).

Items were developed based on principles of classical psychometrics and contemporary theories of organizational cognition. Each dimension is operationalized through multiple observable indicators, enabling the estimation of standardized latent scores (θ) for each participant.

This instrument provides a stable, replicable, and interpretable quantitative foundation for the initial assessment of cognitive–cultural profiles.

2.3.2. Narrative-Based Instrument (NLP–LLM)

The second instrument is based on the collection of open-ended narratives through situational prompts designed to elicit reasoning processes, decision-making patterns, and emotional regulation in complex workplace contexts. This approach enables access to discursive manifestations of situated cognition, beyond declarative responses typical of closed-format instruments.

The prompts are grounded in principles derived from Situational Judgment Tests (SJTs) and the Critical Incident Technique (Flanagan, 1954), encouraging participants to describe real or simulated episodes involving pressure, ambiguity, or goal conflict. This format facilitates the externalization of evaluative criteria, implicit priorities, and cognitive strategies relevant to organizational performance.

The collected narratives are processed using advanced natural language processing (NLP) techniques and large language models (LLMs), selected for their capacity to perform contextual inference, capture long-range semantic dependencies, and sustain coherent reasoning from unstructured text (Devlin et al., 2019; Brown et al., 2020).

In this proof of concept, a transformer-based model with tens of billions of parameters is employed, configured to prioritize inferential consistency, traceability, and interpretive stability.

Model inference parameters (e.g., temperature and sampling controls) were calibrated to reduce stochastic variability and promote explanatory outputs grounded in explicit textual evidence, balancing expressive capacity with methodological control.

From these narratives, the system identifies semantic patterns, argumentative structures, and inferential signals associated with each cognitive–cultural dimension, generating directional classifications, confidence levels, and contextualized evidence fragments.

Unlike closed-response formats, this instrument captures the contextual, dynamic, and situated richness of individual discourse, reducing susceptibility to social desirability bias and strategic response simulation common in self-report measures.

2.4. **Hybrid Scoring and Fusion Mechanism**

The analytical core of the CCFM is based on a controlled fusion mechanism that integrates estimates derived from the psychometric instrument and the automated narrative analysis.

Initially, psychometric data generate a preliminary latent score (θ_{lex}), while the LLM-based module produces a directional inference accompanied by a confidence level (θ_{llm}).

Both sources are combined through an adaptive weighting system, producing a final hybrid score (θ_h), formally defined as:

$$\theta_h = \alpha \cdot \theta_{llm} + (1 - \alpha) \cdot \theta_{lex}$$

where α represents an adaptive weight regulated by the confidence level of the narrative module.

Confidence parameters are estimated based on linguistic and structural indicators, including effective text length, lexical diversity, internal semantic coherence, and thematic stability across the narrative. These factors act as proxies for informational richness and interpretive reliability, reducing the dominance of opaque components while preserving model interpretability (Doshi-Velez & Kim, 2017; Ribeiro et al., 2016).

2.5. Evidence Extraction and Interpretability Layer

To enhance system transparency, the CCFM incorporates an explicit evidence extraction layer. This layer retrieves textual fragments, lexical patterns, and semantic signals that support the generated inferences.

Each dimension includes a structured set of contextualized evidence, accompanied by confidence indicators and interpretive annotations, facilitating human validation and promoting analytical traceability.

2.6. Report Generation and Decision-Support Outputs

Analytical results are integrated into automated reports that include:

i) Standardized dimension scores.

ii) Compatibility diagnostics.

iii) Comparative visualizations.

iv) Contextualized textual evidence

v) Actionable recommendations

These reports function as augmented decision-support artifacts, maintaining the human evaluator as the final agent of interpretation and judgment (Floridi et al., 2018).

2.7. Ethical and Methodological Considerations

The design of the CCFM prioritizes principles of transparency, explainability, and responsible use of artificial intelligence technologies. Given the preliminary nature of this study, direct application in real-world selection contexts is avoided without further empirical validation.

Moreover, the model is conceived as a tool oriented toward human development rather than automated exclusion, promoting an ethical and reflective approach to the analysis of cultural fit.

3. System Architecture and Analytical Pipeline

The Cognitive–Cultural Fit Model (CCFM) is implemented as a modular architecture designed for the multimodal processing of psychological, linguistic, and organizational data. The system integrates specialized components for data acquisition, semantic analysis, hybrid inference, evidence generation, and the production of interpretable reports, following principles of scalable and interpretable analytical system design (Bommasani et al., 2021; Ribeiro et al., 2016).

The architecture is designed around principles of scalability, analytical traceability, and functional separation, enabling the progressive extension of the system to new domains and organizational contexts (Doshi-Velez & Kim, 2017).

3.1. Overall System Architecture

The system is structured into five main functional layers:

3.1.1. **Data Acquisition Layer:** Captures and normalizes data from psychometric instruments, open-ended narratives, and organizational profiles.

3.1.2. **Feature Extraction Layer:** Extracts quantitative and semantic representations from structured and unstructured data.

3.1.3. **Inference Layer:** Integrates multiple data sources through controlled fusion mechanisms.

3.1.4. **Interpretability Layer:** Generates contextualized evidence and traceable explanations.

3.1.5. **Reporting Layer:** Produces automated reports oriented toward decision support.

Each layer operates in a semi-independent manner and communicates through standardized data structures, promoting modularity, maintainability, and extensibility (Bommasani et al., 2021).

3.2. Data Acquisition and Preprocessing Pipeline

3.2.1. Psychometric Data Pipeline

Responses to graded scales undergo initial validation, outlier detection, and statistical normalization. Subsequently, items are aggregated by dimension and transformed into standardized latent estimates, following established psychometric practices (Embretson & Reise, 2000; DeVellis, 2016).

3.2.2. Narrative Data Pipeline

Open-ended narratives are processed through a preprocessing pipeline that includes:

i. Orthographic normalization

ii. Syntactic segmentation

iii. Noise reduction

iv. Tokenization

v. Lemmatization

This pipeline ensures representational consistency prior to deep semantic analysis, in line with standard natural language processing practices (Devlin et al., 2019).

3.3. Semantic and Linguistic Feature Extraction

From the preprocessed text, the system extracts multiple levels of representation.

3.3.1. Lexical–Semantic Indicators

The system employs specialized dictionaries and lexical patterns associated with each cognitive–cultural dimension. These resources enable the identification of explicit signals related to cognitive orientation, emotional regulation, and organizational preferences (Pennebaker et al., 2015).

Indicators include:

i. Normalized frequencies

ii. Semantic polarity

iii. Conceptual density

iv. Thematic co-occurrence

3.3.2. Contextual Embedding Models

In parallel, large-scale language models are used to generate contextual embeddings, which encode semantic, syntactic, and pragmatic information in high-dimensional vector spaces. These representations capture long-range semantic dependencies, discourse relations, and argumentative structures that are not detectable through traditional lexical approaches (Devlin et al., 2019; Brown et al., 2020).

Each narrative is transformed into dense vectors of fixed dimensionality (typically ranging from 768 to several thousand components, depending on the model), where each dimension encodes latent combinations of linguistic and conceptual features learned during training.

These embeddings enable the modeling of similarities, divergences, and relational patterns between individual discourse and organizational profiles, serving as a quantitative foundation for cognitive–cultural analysis.

Based on these representations, the system applies similarity and semantic projection functions to identify regularities in reasoning styles, motivational orientations, and dominant interpretive frameworks. In this way, embeddings support high-level inference regarding latent cognitive patterns and implicit cultural configurations.

3.4. Hybrid Inference Engine

The inference engine constitutes the computational core of the CCFM.

3.4.1. Lexicon-Based Scoring Module

This module computes preliminary scores through the weighted aggregation of lexical indicators by dimension. Each occurrence is evaluated according to its contextual relevance and thematic consistency.

The result is a robust, interpretable, and replicable initial estimate.

3.4.2. LLM-Based Reasoning Module

The LLM module processes full narratives using structured prompts inspired by chain-of-thought and few-shot learning paradigms, adapted to psychological and organizational evaluation contexts (Wei et al., 2022; Brown et al., 2020).

These prompts encourage explicit reasoning, grounding in textual evidence, and internal consistency checks.

Outputs include:

i. Directional classification

ii. Confidence level

iii. Textual evidence

iv. Semantic labels

v. Inferential notes

3.4.3. Fusion Controller

The fusion controller integrates both sources through the adaptive weighting mechanism described in Section 2.4.

This component dynamically regulates the relative influence of each module, mitigating biases derived from incomplete or ambiguous data (Doshi-Velez & Kim, 2017).

3.5. Evidence Generation and Traceability System

To ensure interpretability, the system maintains a structured record of the signals used in each inference.

For each dimension, the system stores:

i. Relevant textual fragments

ii. Lexical activations

iii. LLM-based inferences

iv. Confidence metadata

v. Heuristic annotations

This repository enables post hoc auditing, expert validation, longitudinal analysis, and methodological transparency (Ribeiro et al., 2016).

In this sense, the system operates as an augmented decision-support framework, preserving human interpretive authority (Floridi et al., 2018).

3.6. **Compatibility Computation and Visualization**

Once hybrid scores are generated for each dimension, person–organization compatibility indices are computed using standardized distance and similarity functions in latent vector spaces (Manning et al., 2008).

Let $\theta_p = (\theta_{p1}, \theta_{p2}, \ldots, \theta_{pn})$ represent the individual's score vector, and $\theta_o = (\theta_{o1}, \theta_{o2}, \ldots, \theta_{on})$ represent the organizational profile, where n denotes the number of cognitive–cultural dimensions.

The system employs the following primary metrics:

3.6.1. **Cosine Similarity**

$$\cos(\theta_p, \theta_o) = \frac{\sum_{i=1}^{n} \theta_{p_i} \theta_{o_i}}{\sqrt{\sum_{i=1}^{n} \theta_{p_i}^2} \sqrt{\sum_{i=1}^{n} \theta_{o_i}^2}}$$

This measure evaluates directional alignment between profiles and is particularly useful for identifying similarity in relative cognitive orientation patterns, independent of absolute magnitude.

3.6.2. **Euclidean Distance**

$$d(\theta_p, \theta_o) = \sqrt{\sum_{i=1}^{n} (\theta_{p_i} - \theta_{o_i})^2}$$

This metric captures absolute divergence between profiles, reflecting the total degree of mismatch between individual and organizational configurations.

Both values are subsequently transformed into normalized interpretive indices, enabling their integration into compatibility scales that are easily communicable to non-technical users.

For analytical and applied purposes, results are visualized through:

- Dimension-level comparison matrices
- Multidimensional radar charts
- Global compatibility indices
- Longitudinal profiles (in extended system versions)

These representations facilitate rapid identification of alignment zones, potential friction, and trajectories of cultural adaptation over time.

3.7.Automated Report Generation Engine

The final module integrates all analytical components into structured, automatically generated documents, ensuring methodological coherence, interpretive traceability, and reproducibility.

The engine produces:

- Reports in Markdown format for technical transparency
- Automatic export to DOCX via Pandoc for institutional integration
- Traceability metadata, including model versions, analytical parameters, and timestamps
- Automated versioning mechanisms for auditing and replication

Each report constitutes a reproducible and verifiable artifact, designed for use by HR teams, organizational psychologists, and technical leaders, as well as for scientific documentation and longitudinal evaluation.

3.8. **System Workflow Summary**

The full system workflow can be summarized as follows:

i. Data collection

ii. Preprocessing

iii. Feature extraction

iv. Lexical and contextual inference

v. Controlled fusion

vi. Evidence generation

vii. Compatibility computation

viii. Report generation

This pipeline ensures methodological consistency, interpretability, and adaptability.

4. Results and Proof of Concept

To illustrate the operational functioning of the Cognitive–Cultural Fit Model (CCFM), a proof of concept was developed based on a simulated scenario integrating synthetic individual profiles, reference organizational parameters, and narratives generated through the hybrid instrument.

The objective of this demonstration is to evaluate the system's capacity to produce interpretable, coherent, and actionable diagnostics in person–organization evaluation contexts.

4.1.Case Study Configuration

The analyzed case corresponds to a professional profile with high technical competence in software development environments, characterized by a preference for goal orientation, autonomy, and flexibility, evaluated against an organization defined by formal structures, high operational demands, and strong results orientation.

The evaluation integrated:

i. Standardized psychometric results
ii. Narrative responses to situational prompts
iii. Semantic analysis using language models
iv. Integration through the controlled fusion mechanism

Based on these inputs, latent scores (θ) were generated across six cognitive–cultural dimensions.

4.2. Quantitative Results

4.2.1. Overall Compatibility

The system produced the following global indicators:

- Overall fit level: Low
- Mean $|\Delta\theta|$: 1.01
- Median $|\Delta\theta|$: 1.09
- Interpretive compatibility: 33%

These results indicate a low overall level of alignment between the individual profile and the organizational environment, suggesting a high risk of cultural friction and sustained adaptation difficulties.

4.2.2. Dimension-Level Analysis

Table 1 presents comparative results across dimensions.

Table 1. Person–Organization Compatibility by Dimension

Comparative Matrix (Quick View)

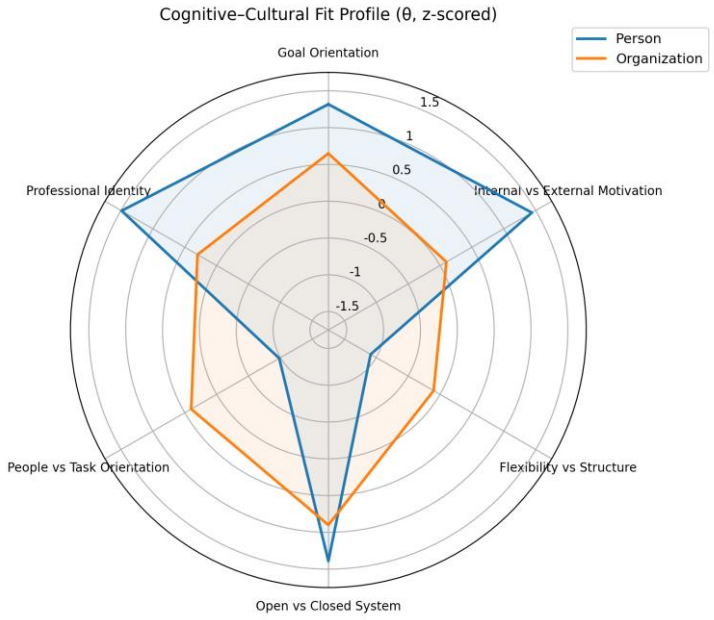
Dimension	Person θ	Org θ	Δθ (P−O)	|Δθ|	Compatibilidad interpretativa	Level
Goal Orientation	+1.32	+0.65	+0.67	0.67	56%	Moderate-to-low compatibility
Internal vs external motivation	+1.44	+0.10	+1.34	1.34	11%	Low compatibility
Flexibility vs structure	-1.09	-0.10	-0.99	0.99	34%	Low compatibility
Open vs Closed System	+1.39	+0.90	+0.49	0.49	67%	Moderate compatibility
People vs task orientation	-0.98	+0.40	-1.38	1.38	8%	Low compatibility
Professional identity	+1.49	+0.30	+1.19	1.19	21%	Low compatibility

Partial alignment is observed in goal orientation and organizational openness. However, most dimensions exhibit substantial divergence, particularly in motivation, structural regulation, and socioemotional orientation.

4.2.3. Graphical Representation

Figure 1 presents the comparative cognitive–cultural profile of the individual and the organization.

Figure 1. Cognitive–Cultural Profile (Radar Chart)



The visualization reveals systematic asymmetries between both profiles, reinforcing the quantitative diagnosis and facilitating intuitive interpretation.

4.3. Narrative Evidence and Latent Signals

The system generates textual evidence associated with each dimension, enabling traceability of the interpretive origin of the scores.

Example: Goal Orientation

- **Evidence excerpt:** "The priority was the outcome: we released a functional version to avoid blocking deployment."

Inferred parameters:

- **Dominant pole**: Goal-Oriented
- **Confidence:** 0.85
- **Observation:** Low lexical activation, high inferential weighting

This type of evidence establishes a direct link between numerical indicators and concrete linguistic manifestations, strengthening interpretability and analytical transparency.

4.4. Integrated Interpretation

The results indicate a low overall compatibility between the individual profile and the simulated organizational context.

Although partial alignment is observed in goal orientation (Goal Orientation) and organizational openness (Open vs Closed System), significant divergences emerge in motivation (Internal vs External Motivation), Structural preference (Flexibility vs Structure) and Relational orientation (People vs Task Orientation). These divergences form a pattern of systemic misalignment

This profile suggests:

- High risk of operational friction
- Potential motivational disengagement
- Medium-term integration difficulties
- Increased vulnerability under high-pressure conditions

From an applied perspective, the CCFM demonstrates the ability to detect early signals of potential incompatibility, enabling preventive intervention strategies.

4.5. Prescriptive Recommendations

Based on the diagnostic outputs, the system generated dimension-specific recommendations, including:

- Structured onboarding strategies
- Role design adjustments

- Targeted coaching interventions
- Early indicators of cultural risk

This approach moves beyond static classification, positioning the model as a development-oriented and adaptive decision-support system.

4.6. **Automated Report Generation**

Based on the analytical results, the system generated a structured report including:

- Executive summary
- Comparative matrix
- Radar visualization
- Dimension-level diagnostics
- Contextual evidence
- Actionable recommendations

The report was exported in both Markdown and DOCX formats, facilitating integration into organizational processes such as selection, development, and cultural monitoring.

4.7. Case Summary

The proof of concept demonstrates that the CCFM:

- Integrates quantitative and narrative data
- Produces explainable diagnostics
- Generates verifiable evidence
- Provides prescriptive guidance

- Maintains theoretical coherence

These results support the functional viability of the model as a cognitive–cultural assessment system in organizational contexts.

4.8. Limitations of the Case

Given the synthetic nature of the scenario, the results should not be interpreted as definitive empirical evidence.

However, the case provides an initial functional validation of the model and establishes a foundation for future longitudinal evaluations in real organizational settings.

5. Discussion, Implications, and Future Research

5.1. Discussion of Results

The results obtained through the Cognitive–Cultural Fit Model (CCFM) highlight the relevance of cognitive–cultural alignment as a structural factor in performance, integration, and the sustainability of the person–organization relationship.

The case analysis demonstrated that, even among profiles with high technical competence, the presence of systematic misalignments across key dimensions may generate latent risks of friction, motivational strain, and early turnover. These findings reinforce the literature on person–organization fit while extending its scope through the incorporation of linguistic evidence and contextual semantic analysis.

Moreover, the partial convergence observed across certain dimensions suggests that adaptation processes do not depend exclusively on overall compatibility levels, but rather on specific configurations of strengths and cultural tensions. In this sense, the CCFM enables a

shift beyond binary “fit/misfit” evaluations, providing multidimensional profiles that are more informative and dynamically interpretable.

5.2.Theoretical Implications

From a theoretical perspective, the CCFM contributes to the integration of frameworks from cognitive science, organizational psychology, and applied artificial intelligence.

First, the model operationalizes constructs related to situated cognition and cognitive identity in workplace contexts, translating them into measurable and comparable dimensions. This enables a more dynamic, contextualized, and functional conceptualization of cultural fit.

Second, the controlled fusion strategy represents a relevant methodological contribution, demonstrating that psychometric and linguistic data can be integrated without sacrificing interpretability or analytical rigor. This approach addresses long-standing limitations of models based exclusively on self-report measures or opaque machine learning systems.

Finally, the CCFM proposes a replicable and adaptable hybrid architecture, strengthening the dialogue between psychological theory, computational methodologies, and organizational applications.

5.3.Practical Implications

From an applied perspective, the results highlight the potential of the CCFM as an advanced decision-support tool in organizational contexts.

In selection processes, the model enables early identification of cultural misalignment risks, complementing traditional assessments of technical competencies and soft skills. In

onboarding and development contexts, it facilitates the personalization of training and support interventions.

Furthermore, the generation of explainable reports and actionable recommendations promotes an ethical and developmental use of results, oriented toward human growth rather than systematic exclusion.

From a talent management perspective, the CCFM contributes to reducing costs associated with turnover, internal conflict, and disengagement by supporting evidence-based, multidimensional decision-making.

5.4. Study Limitations

This study presents several limitations that should be considered when interpreting the results.

First, the study is based on a synthetic scenario, which limits the empirical generalizability of the findings. Although the case supports functional validation of the system, longitudinal studies in real organizational contexts are required to assess its long-term impact.

Second, the performance of the linguistic component depends on the quality, length, and authenticity of participant-generated narratives, which may introduce uncontrolled variability.

Additionally, although the controlled fusion mechanism mitigates algorithmic bias, the partial reliance on large-scale language models trained on extensive corpora entails inherent risks of cultural and contextual bias.

5.5. Future Research Directions

Based on the present findings, several promising directions for future research are identified.

First, empirical validation of the model through longitudinal studies in real organizational settings is required, incorporating performance metrics, retention indicators, and measures of employee well-being.

Second, future research may explore the expansion of the CCFM dimensional framework by incorporating variables related to leadership, cultural diversity, and intergroup dynamics.

Additionally, further development of specialized conversational agents capable of conducting adaptive, real-time evaluations represents a relevant line of advancement, particularly in the context of continuous feedback systems.

Finally, the integration of the CCFM into AI-based talent management platforms could foster human-centered, augmented decision ecosystems, enhancing organizational adaptability and developmental outcomes.

5.6. **General Conclusion**

The Cognitive–Cultural Fit Model represents an innovative approach to the analysis of person–organization alignment, combining solid theoretical foundations with advanced linguistic and psychometric analytical techniques.

By addressing the limitations of traditional approaches and providing explainable diagnostics, contextual evidence, and actionable recommendations, the model contributes to the development of more transparent, fair, and development-oriented evaluation systems.

Overall, this work demonstrates the potential of integrating cognitive science, organizational psychology, and artificial intelligence as a pathway to transforming human evaluation processes in complex organizational environments.

References

Barsalou, L. W. (2008). Grounded cognition. Annual Review of Psychology, 59, 617–645.

Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., … Liang, P. (2021). On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258.

Brown, J. S., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. Educational Researcher, 18(1), 32–42.

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., … Amodei, D. (2020). Language models are few-shot learners. Advances in Neural Information Processing Systems, 33, 1877–1901.

Cable, D. M., & Judge, T. A. (1996). Person–organization fit, job choice decisions, and organizational entry. Organizational Behavior and Human Decision Processes, 67(3), 294–311.

Clark, A. (1997). Being there: Putting brain, body, and world together again. MIT Press.

Creswell, J. W. (2014). Research design: Qualitative, quantitative, and mixed methods approaches (4th ed.). Sage.

Denison, D. R. (1990). Corporate culture and organizational effectiveness. Wiley.

DeVellis, R. F. (2016). Scale development: Theory and applications (4th ed.). Sage.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT, 4171–4186.

Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608.

Embretson, S. E., & Reise, S. P. (2000). Item response theory for psychologists. Lawrence Erlbaum.

Flanagan, J. C. (1954). The critical incident technique. Psychological Bulletin, 51(4), 327–358.

Hofstede, G. (2001). Culture's consequences: Comparing values, behaviors, institutions, and organizations across nations (2nd ed.). Sage.

Hofstede, G., Hofstede, G. J., & Minkov, M. (2010). Cultures and organizations: Software of the mind (3rd ed.). McGraw-Hill.

Kristof-Brown, A. L. (1996). Person–organization fit: An integrative review. Journal of Vocational Behavior, 49(1), 1–49.

Kristof-Brown, A. L., Zimmerman, R. D., & Johnson, E. C. (2005). Consequences of individuals' fit at work. Personnel Psychology, 58(2), 281–342.

Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research. Journal of Applied Psychology, 88(5), 879–903.

Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. Proceedings of the ACM SIGKDD, 1135–1144.

Schachner, M., Ardag, M. M., Holtz, P., Großer, J., Hartz, C., van Herk, H., & Dobewall, H. (2024). Extracting organizational culture from text: The development and validation of a

theory-driven tool for digital data. *European Journal of Work and Organizational Psychology, 33*(5), 571–582. https://doi.org/10.1080/1359432X.2024.2360225

Schein, E. H. (2010). Organizational culture and leadership (4th ed.). Jossey-Bass.